
Electromagnetic Interference in Analogue and Digital Systems

An Open-Source Summary

Thomas Debelle



October 10, 2025

Contents

1 Terminology	1
1.1 Basic Scheme	2
1.1.1 Common mode and Differential mode	3
1.1.2 Size of a system	3
2 Coupling Mechanisms	3
2.1 Transition between conductor and dielectric	4
2.1.1 Transition between medium	5
2.1.2 Link with the term capacity	5
2.2 Combined coupling	5
3 Multi-conductor Transmission Lines	6
3.1 Characteristic impedance matrix	9
3.2 Crosstalk	9
3.2.1 Special case: homogenous	9
3.2.2 Transfer impedance	10
4 Shielding	11
5 License	13
5.1 Modification	13

1 Terminology

- **Electromagnetic Compatibility (EMC):** the fact that devices can work normally in the neighborhood of other. Capability of a device, apparatus or system to function properly (with a pre-defined margin) in its electromagnetic environment (given in advance) without generating intolerable disturbing signals in this environment
 - **EM Emission:** emission; to disturb. within and between systems
 - **EM Susceptibility:** to be disturbed. within and between systems
 - Everything is a probability and we must take margins to ensure the proper function of a circuit.
- **Electromagnetic Interference (EMI):** the fact that device can be influenced by other. Degradation of the functioning of a device, apparatus or system as a consequence of a disturbing signal; degradation = intolerable behavior, hard to define.
 - This is more and more problematic as:
 - * Density increase; speed increase
 - * More broadband
 - * Small and High amplitude

This course is all about understanding the fundamental of what makes a circuit and electricity flow. We ground for safety and no current flows in GND in a household. A coaxial is not shielded ! When nothing powers it, it doesn't act as a shielding, only after we power it it will look as shielded.

General rule for disturbance and interference:

- Disturbance: signal causing the problem
- Interference: problem

One thing to always consider is to also check for possible interferences outside the working band as external phenomena happening at higher or lower frequencies can be perceived in the circuit.

1.1 Basic Scheme

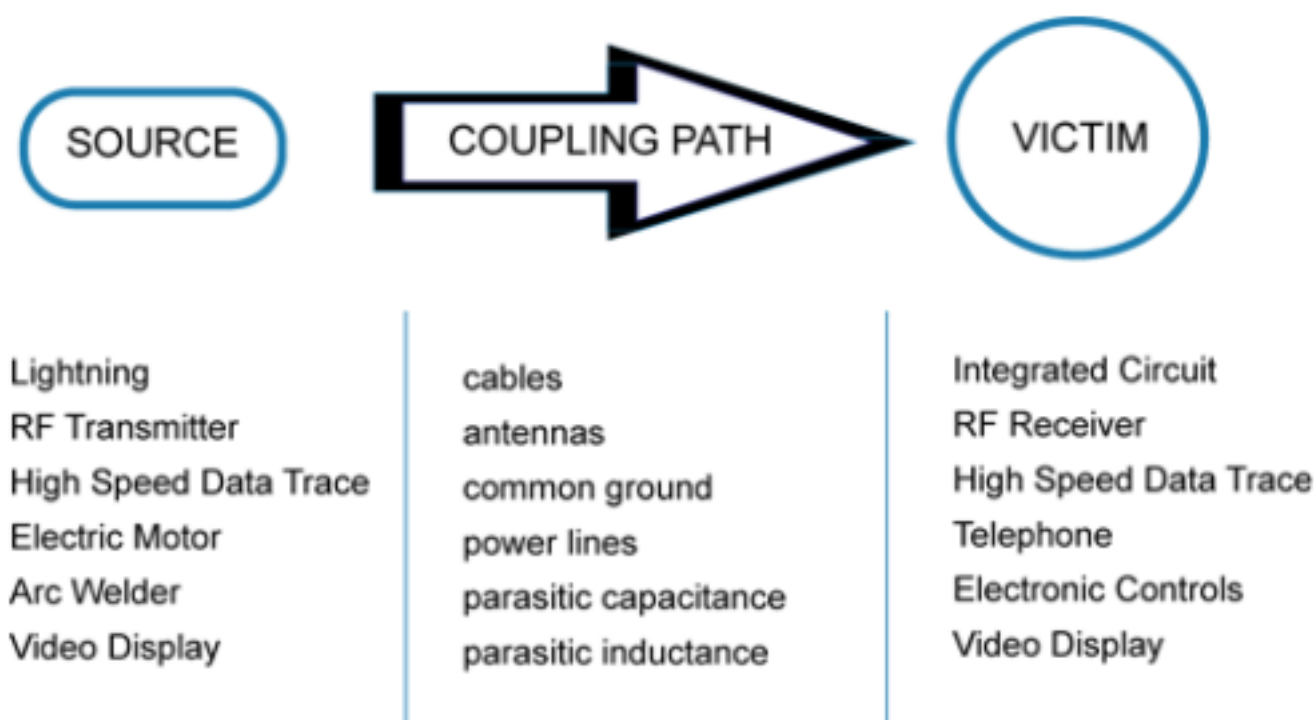


Figure 1: Basic scheme

Important to note that there is not just 1 coupling path. It can be inductive, capacitive, ... but can be also conducted through the wires !

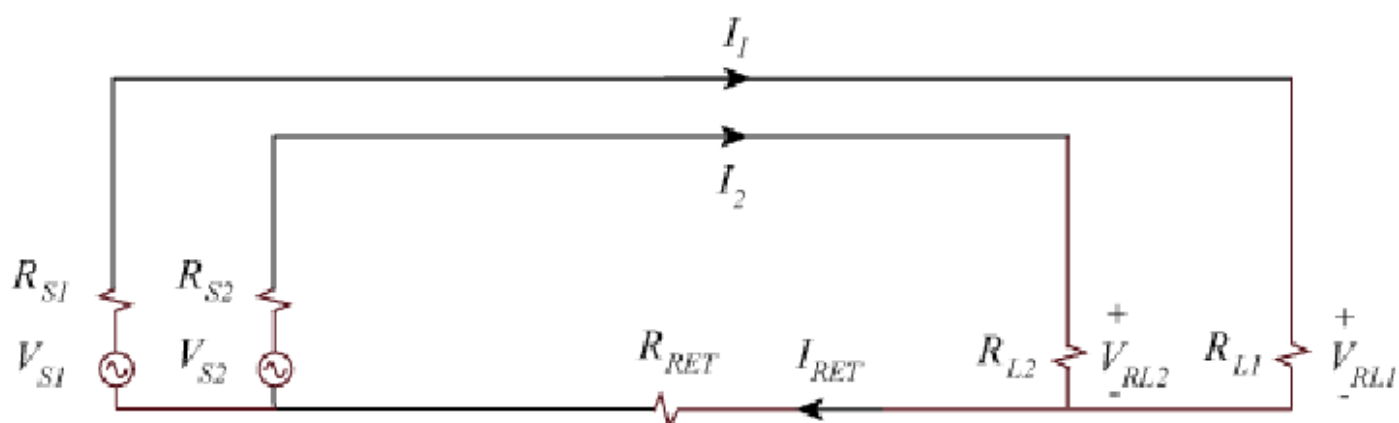


Figure 2: Common impedance coupling

On top of this we can have some common impedance coupling where the length and slight different in material results in differences in the signal.

1.1.1 Common mode and Differential mode

They are both important in EM but generally, we are interested in the DM as no current is going through the ground. Common mode is problematic due to its fact the signal go to ground. This could also create an unexpected path that could damage components.

1.1.2 Size of a system

It is always related to the wavelength λ .

- $< \lambda/10$: Small: quasi-stationary
- $> \lambda/10$: Large: wave behavior

For the near and far-field we use:

- $< \lambda/2\pi$: near-field
- $> \lambda/2\pi$: far-field

For antenna, we must also take into account the directivity with 10λ and $2D^2/\lambda$.

Another reminder is the fact that for a signal in dB, we use $20\log(X(y))$ but if it is the power, we use $10\log(X(y)_{power})$ as it is a quadratic value.

2 Coupling Mechanisms

Note

Physical meaning: the notions potential and voltage actually **ONLY EXIST AT DC**.

This can be proven with the basic Maxwell equation:

$$\nabla \times E = -\frac{dB}{dt}$$

If we are at DC, no change of rate respecting the time so $dB/dt = 0$. We also know that $\nabla \cdot \nabla \times V = 0$ by definition. This leads us to $E = -\nabla\phi$ with the $-$ as a convention so we go from high to low potential. Finally, we can transform this equation into an integral equation:

$$\int_1^2 E dl = \phi_1 - \phi_2$$

In the case of an AC signal where the derivate is no longer null we have:

$$\int \nabla \times E dS = \int -\frac{dB}{dt} dS \quad \text{with } \int \nabla \times E dS = \oint E dC \text{ the contour} \quad (1)$$

$$\oint E dC = \int -\frac{dB}{dt} dS \quad \text{with } B = \frac{\phi}{dS} \quad (2)$$

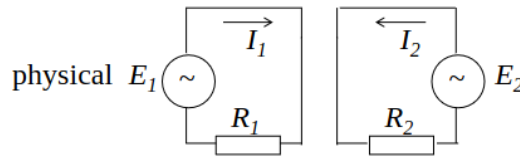
$$\oint E dC + \frac{d\phi}{dt} = 0 \quad (3)$$

In other words, the sum of the voltage drop over a closed loop is not equal to 0! This contradicts the KCL.

Which means that when we want to model any RC circuit we will find:

$$\begin{cases} \oint E dC &= -V + RI \\ \frac{d\phi}{dt} &= \frac{dLI}{dt} = L \frac{dI}{dt} \end{cases}$$

Where we have first the voltage drop and then the extra term cause by the loop and the creation of a surface where a magnetic flux is going through. In short, we must model a simple resistive circuit with an extra inductance to keep it correct with Maxwell equation.



Choose direction in which to go through the loop.

$$\begin{aligned} \text{loop 1: } \oint_{C_1} E dC_1 &= -E_1 + R_1 I_1 & \text{loop 2: } \oint_{C_2} E dC_2 &= -E_2 + R_2 I_2 \\ \frac{d\phi_1}{dt} &= \frac{d(L_{11}I_1 + L_{12}I_2)}{dt} & \frac{d\phi_2}{dt} &= \frac{d(L_{21}I_1 + L_{22}I_2)}{dt} \\ &= L_{11} \frac{dI_1}{dt} + L_{12} \frac{dI_2}{dt} & &= L_{21} \frac{dI_1}{dt} + L_{22} \frac{dI_2}{dt} \end{aligned}$$

Figure 3: Mutual coupling

Each pair of loop as to be modeled as a **transformer**!

2.1 Transition between conductor and dielectric

Circuit theory is a simple first order approximation of field theory. When we have a dielectric/vacuum on top of a conductor, we know that the netto charge in conductor is null.

$$\nabla \times H_2 = (j\omega\epsilon_2 + \sigma_2)E_2 \text{ with } J_2 = \sigma_2 E_2$$

And as used before $\nabla \cdot \nabla \times V = 0$ and so the divergence of the current. Which means what comes in comes out ! The net is null.

2.1.1 Transition between medium

Now, if we inspect a small loop going around the first and second medium, we can find:

$$\int_V \nabla(j\omega\varepsilon + \sigma)EdV = \oint_S (j\omega\varepsilon + \sigma)EdS = 0$$

There is a current flowing to the surface of the conductor that is related to the field outside the conductor:

$$J_{2n} = \frac{\sigma_2 j\omega\varepsilon_1}{(j\omega\varepsilon_2 + \sigma_2)} E_{1n}$$

And at DC and with a perfect conductor ($\sigma_2 \rightarrow \infty$), we have $J_{2n} = j\omega\varepsilon_1 E_{1n}$. This appears immediately after switching it on, this charge ensures the correct voltage drop outside. And later inside, the electrical field is created for the current flow.

2.1.2 Link with the term capacity

This phenomena is the same that govern the capacity in its strict definition. If we have an area A at a distance d from each other with a voltage potential of V and permittivity of ε . We have:

$$J_{2n} = j\omega\varepsilon_1 E_{1n} \longrightarrow \underbrace{AJ_{2n}}_I = j\omega \underbrace{\frac{\varepsilon_1 A}{d}}_C \underbrace{dE_{1n}}_V$$

This can be applied to **any pair of conductors**, which is contrary to the current law of Kirchoff. We must model it as a RC parallel network that appears at AC.

2.2 Combined coupling

Capacitive and inductive couplings are **first order correction**. Any one port system can be modeled with a Maclaren series:

$$Z(\omega) = R + A_1\omega + A_2\omega^2 + A_3\omega^3 + \dots$$

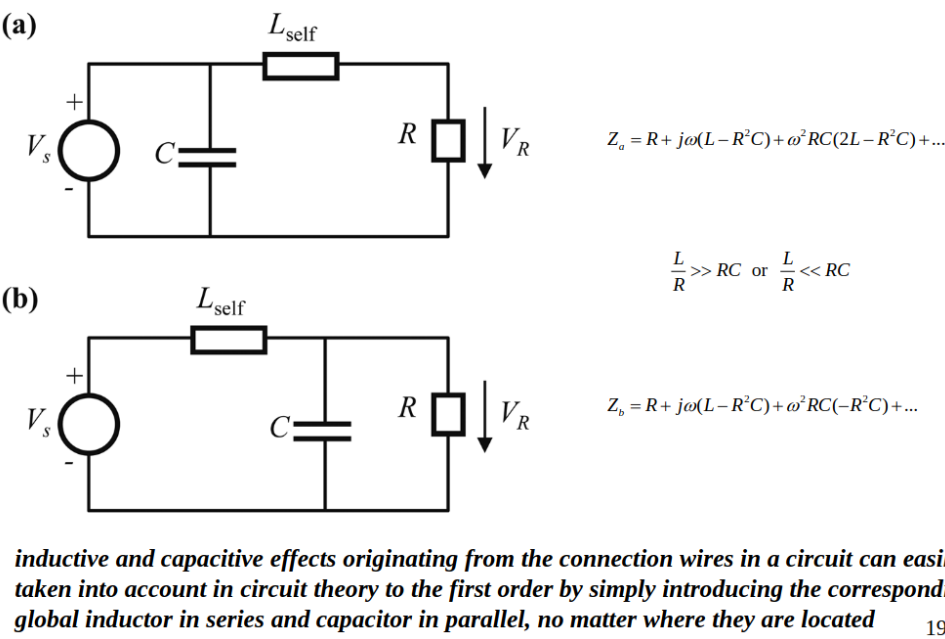


Figure 4: Combined coupling

Both circuits are equivalent but will have different values. We can then split-up the two networks and boom we have the same second order term by choosing wisely our split. This will lead to $L_{2R}C_{2R} = L_{2s}C_{2s}$ for an equal split.

Also, important to note that this combined coupling will change depending on the actual wiring and physical constraint which is sometimes forgotten about.

Table 1: Consequences of higher order phenomena

Characteristics	Circuit theory	Reality
Conductors	Ideal just transport current and voltage	It is a component itself with non-idealities
Shape and implementation	No impact, just the components matter	At higher speed shape matters a lot
Components	Localized, discrete	Distributed not discrete

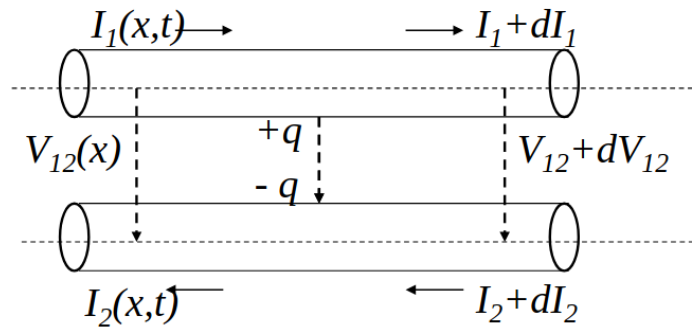
If we do not take care of this, we have lots of problem especially at higher frequencies as such effect because increasingly important. We have to switch gears and really ask ourselves what simplification does circuit theory.

- inductive coupling: make current loop as small as possible
- capacitive coupling: distance as short as possible

3 Multi-conductor Transmission Lines

This is a generalization of the two conductors case where we witness some C and L coupling between conductors of length $\gg \lambda$. This could be applied to shorter wires but this theory will be equal to circuit theory but with extra work and computation.

If we look transversal, the cross section is much smaller which leads to quasi-static (quasi TEM) and so possible to introduce L and C for correction.



$$\oint_C \mathbf{E} d\mathbf{C} + j\omega\phi = 0$$

$$\mathbf{J}_n = j\omega\epsilon_{\text{medium}} \mathbf{E}_n$$

Figure 5: L and C coupling

If we look in telecommunication cables such as telephone line or ethernet cable, we will see twister pairs, shielding, ... this is a first hint to answer the question “how to reduce coupling”. This same principle is also found in microstrip line. The strongest coupling is the horizontal one between the microstrip but also some with the ground plate.

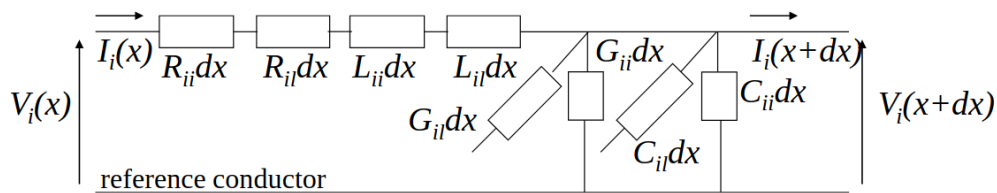
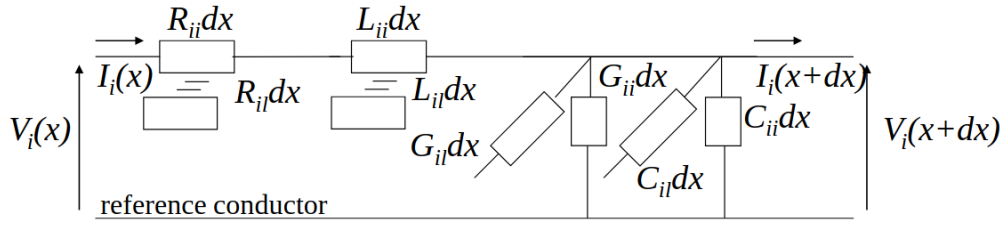


Figure 6: Transmission line equation (frequency domain)

Each components model the effects within a TL and with the other TL. Distributed (dx) components determined via electrostatics and magnetostatics in cross section. On small dx we could use KCL. This models is **only valid if cross-section** $\ll \lambda$. (why?)



$$\left\{ \begin{array}{l} V_i(x+dx) = V_i(x) + \frac{dV_i(x)}{dx} dx + \dots \quad \text{en} \quad I_i(x+dx) = I_i(x) + \frac{dI_i(x)}{dx} dx + \dots \\ \frac{dV_i(x)}{dx} = -\sum_{l=1}^n R_{il} I_l(x) - j\omega \sum_{l=1}^n L_{il} I_l(x) \\ \frac{dI_i(x)}{dx} = -G_{ii} V_i(x) - \sum_{l=1, l \neq i}^n G_{il} (V_l(x) - V_i(x)) - j\omega C_{ii} V_i(x) - j\omega \sum_{l=1, l \neq i}^n C_{il} (V_l(x) - V_i(x)) \end{array} \right.$$

two coupled sets of 1st order differential equations

Figure 7: More explicit model

If we lump everything in one single vector component, we can obtain a matrix representation with:

$$Rdx + j\omega Ldx = Zdx \qquad Gdx + j\omega Cdx = Ydx \qquad (4)$$

and

$$\begin{cases} \frac{dV(x)}{dx} = -\mathbf{Z}\mathbf{I}(x) \\ \frac{dI(x)}{dx} = -\mathbf{Y}\mathbf{V}(x) \end{cases}$$

If the matrix becomes of dimension 1, we are in the well-known case of a transmission line. This approach is simply a generalization of it. The matrices are like:

$$\mathbf{Z} = \begin{bmatrix} Z_{11} & Z_{12} & \dots & Z_{1n} \\ Z_{21} & Z_{22} & \dots & Z_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ Z_{n1} & Z_{n2} & \dots & Z_{nn} \end{bmatrix} \qquad \mathbf{Y} = \begin{bmatrix} \sum_{i=1}^n Y_{1i} & -Y_{12} & \dots & -Y_{1n} \\ -Y_{21} & \sum_{i=1}^n Y_{2i} & \dots & -Y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -Y_{n1} & -Y_{n2} & \dots & \sum_{i=1}^n Y_{ni} \end{bmatrix} \qquad (5)$$

If we assume $V(x) = V^+ e^{-\gamma x}$ and $I(x) = I^+ e^{-\gamma x}$, if we use the second derivate of the equation, we get:

$$(\gamma^2 U - \mathbf{Z}\mathbf{Y})V^+ = 0 \qquad (\gamma^2 U - \mathbf{Y}\mathbf{Z})I^+ = 0$$

Where U is the modal transformation matrix related. The solution is thus given by the eigenvalues γ_i and the eigenmodes V_i^+ ; I_i^+ . If we look at the column of the current and voltage, they each represent 1 mode-1 base of the system. The matrix is full rank so each mode is orthogonal to the others.

$$I_m = [I_{m1} \quad I_{m2} \quad \dots \quad I_{mn}] \quad V_m = [V_{m1} \quad V_{m2} \quad \dots \quad V_{mn}]$$

3.1 Characteristic impedance matrix

We can derivate the equations we found and injecting the new notation of current and voltage which yield:

$$V_{mi} = \gamma_i^{-1} Z I_{mi} \quad V_{mi} = \gamma_i Y^{-1} I_{mi}$$

We can the repeat for every entry of the matrix and by sorting the terms properly for γ which will become a diagonal matrix $[\gamma]$:

$$V_m = (Z I_m)[\gamma]^{-1} = (Y^{-1} I_m)[\gamma] \quad I_m = Z^{-1} I_m[\gamma] = Y V_m[\gamma]^{-1}$$

$$V_m = Z_c I_m \Rightarrow Z_c = V_m I_m^{-1} = (Z I_m)[\gamma]^{-1} I_m^{-1} = (Y^{-1} I_m)[\gamma] I_m^{-1} \\ = V_m[\gamma]^{-1} V_m^{-1} Z = V_m[\gamma] V_m^{-1} Y^{-1}$$

Figure 8: Characteristic impedance matrix

3.2 Crosstalk

Definition

Crosstalk is any phenomenon by which a signal transmitted on one circuit or channel of a transmission system creates an undesired effect in another circuit or channel. Crosstalk is usually caused by undesired capacitive, inductive, or conductive coupling from one circuit or channel to another

So if we end the multi conductor TL with the impedance matrix it can remove the crosstalk.

The main takeaway from this is that, L and C coupling are unavoidable and it is a **physical** phenomena. But cross-talk is a result of this but **can be avoided** if we are putting data on one mode only.

3.2.1 Special case: homogenous

If the medium is homogeneous, like in power distribution line for example, $\gamma_i \equiv \gamma_{\text{medium}}$. This means that:

$$Z = \gamma_{\text{medium}}^2 Y^{-1}$$

This yields that:

$$j\omega L = (j\omega \sqrt{\mu_{\text{medium}} \varepsilon_{\text{medium}}})^2 (j\omega C)^{-1} \quad L = (\mu_{\text{medium}} \varepsilon_{\text{medium}}) C^{-1}$$

This makes matching possible as it will create a real-valued $Y_c = \sqrt{\mu_{\text{medium}} \varepsilon_{\text{medium}}}^{-1} C$. And so each components is a real valued resistor.

3.2.2 Transfer impedance

This concept indicates us how signal on the outside will influence the signal inside. Basic insight is to make the cable as short as possible and the connection as good as possible to avoid excess inductive coupling.

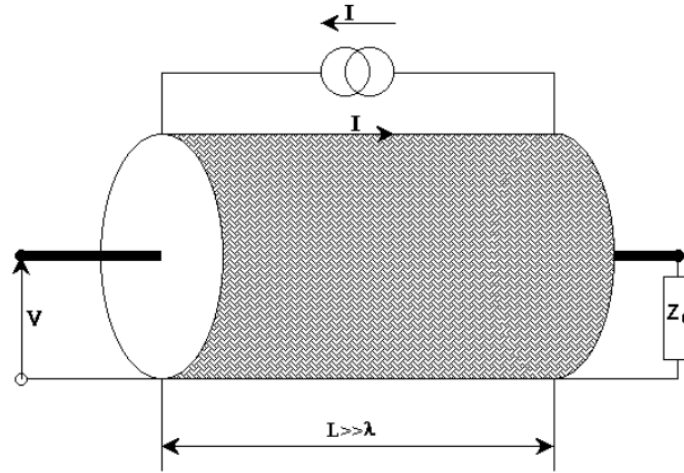
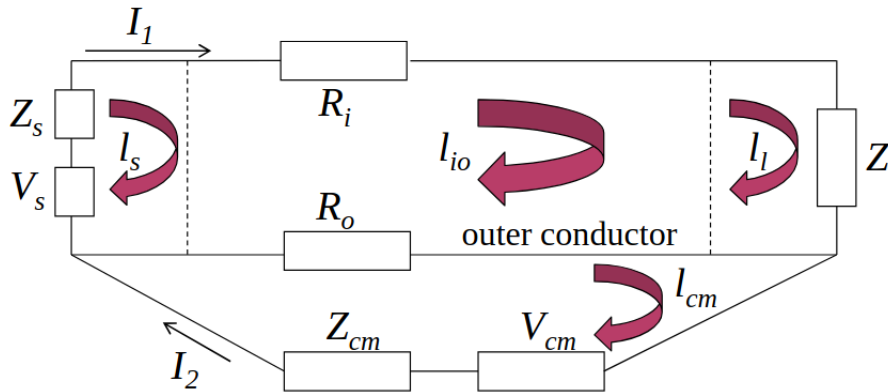


Figure 9: Transfer impedance



$$-V_s + Z_s I_1 + R_i I_1 + Z_l I_1 + R_o (I_1 - I_2) + j\omega L_1 I_1 - j\omega M I_2 = 0$$

$$-V_{cm} + Z_{cm} I_2 + R_o (I_2 - I_1) + j\omega L_2 I_2 - j\omega M I_1 = 0$$

Figure 10: At low frequencies

3.2.2.1 Low frequencies On the left we have the sources, then the cable itself with R_i and finally the load that is not active. Finally, we have the return path on the outer conductor. The extra loop at the bottom represents the common mode that is disturbing th signal, this will create a signal inside.

We have then the inner and outer loop equations. The L and M represents the self-inductance and mutual inductance between the loops. It appears due to reciprocity of Maxwell's equation and causes L coupling.

We are interested in the effects of the cable not the connections. For this, we will split the inner loop into the source, load and cable.

$$\begin{aligned}
& -V_s + Z_s I_1 + R_i I_1 + Z_l I_1 + R_o I_1 - R_c I_2 \\
& + j\omega(L_{ss} + L_{cc} + L_{ll} - M_{sc} - M_{sl} - M_{cs} - M_{cl} - M_{ls} - M_{lc}) I_1 \\
& - j\omega(M_{s2} + M_{c2} + M_{l2}) I_2 = 0 \\
& -V_{cm} + Z_{cm} I_2 + R_o I_2 - R_o I_1 + j\omega L_2 I_2 - j\omega(M_{2s} + M_{2c} + M_{2l}) I_1 = 0
\end{aligned}$$

effect of I_2 on coax:

$$Z_T = R_o + j\omega M_{c2}$$

Figure 11: Splitting of cable

3.2.2.2 High frequencies Cause of the skin depth effect, an exponential decay of exterior signal will be witnessed in the outer conductor. Ideally the characteristic impedance should be 0 if the shielding is thick enough.

But if we look inside standard commercial coax cable, we will see a *woven metal* which will allow for flexibility. This will reduce its performance as depicted here:

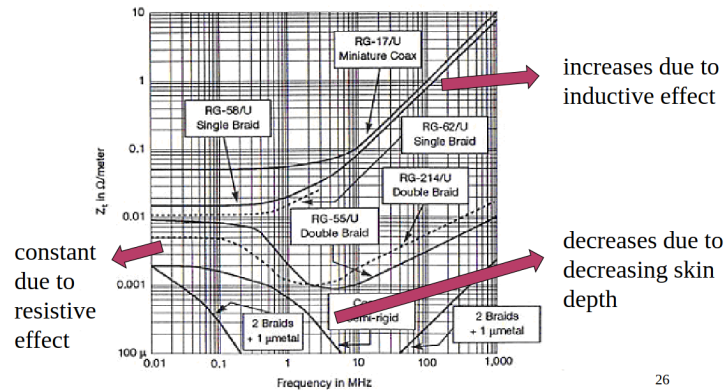


Figure 12: Plot of the performance of various shielding

For high frequencies, the woven metal will have holes and so no skin-depth. Full copper will increase the skin-depth effect as the frequency increases. On the graph, the *mu-metal* is nickel/iron alloys with the permeability μ .

4 Shielding

Definition

Shielding = to diminish the electromagnetic field at the location of the susceptor by placing a physical barrier, in most cases consisting of a conducting material, between emitter and susceptor

We must shield from $\vec{E}$ and $\vec{H}$, those are 2 different cases. For plane wave, achieving one will achieve the other as their fields are orthogonal and their ratio equal to the medium constant.

We can either shield a device or shield the source of disruption. A general rule is that the higher the σ , the better is the shielding. BUT, the topology can impact it drastically.

Table 2: Typical materials for shielding

Metal	Conductivity - σ	Permeability - μ
Silver	1.05	1
Copper	1	1
Gold	0.7	1
Aluminum	0.61	1
Zinc	0.29	1
Brass	0.26	1
Nickel	0.2	1
Iron	0.17	1000
Tin	0.15	1
Steel	0.1	1000
Hypernick	0.06	80000
Monel	0.04	1
Mu-Metal	0.03	80000
Stainless Steel	0.02	1000

5 License

This work is licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License (CC BY-NC-SA 4.0).

You are free to:

- **Share** — copy and redistribute the material in any medium or format.
- **Adapt** — remix, transform, and build upon the material.

Under the following terms:

- **Attribution** — You must give appropriate credit, provide a link to the license, and indicate if changes were made. You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use.
- **NonCommercial** — You may not use the material for commercial purposes.
- **ShareAlike** — If you remix, transform, or build upon the material, you must distribute your contributions under the same license as the original.
- **No additional restrictions** — You may not apply legal terms or technological measures that legally restrict others from doing anything the license permits.

For the full legal text of the license, please visit: <https://creativecommons.org/licenses/by-nc-sa/4.0/legalcode>

5.1 Modification

To contribute to this work or any other one from this project please find more information at the Github repository.

© 2025 Authors of the Summary, Professors of the Course and possible book's authors. Some Rights Reserved.